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CELBC710 - Computational Intelligence Laboratory | Semester VII | AY 2026-2027

Experiments 9 and 10

System Integration & Deployment, and Evaluation, Responsible AI & Documentation

Final Project Report: SymbioForge, an Intelligent Clinical Decision Support System for Cardiovascular Disease

Tools: Python, XGBoost, NumPy/PyTorch, Pydantic, Streamlit, Plotly, pytest

1. Objective and Summary

To integrate the eight earlier experiments (knowledge representation, rule-based expert system, state-space search, data preparation, feature engineering, machine learning, deep learning and explainable AI) into one working prototype, and to evaluate it for performance, reliability, responsible use and deployment readiness.

SymbioForge is not a single classifier. It is a team of independent AI workflows under one orchestrator. A user describes their symptoms in a guided chat; each workflow examines the answers from its own angle (statistical model, deep network, expert rules, explanation, trust, what-if sensitivity, test planning); the orchestrator merges the evidence into a patient-level overview that explains what stands out, how sure the system is, and what to do next. A reviewer-facing view exposes the machinery behind it.

Decision support only, not a diagnosis. The prototype is trained on a public research dataset and is not clinically validated.

2. System Overview: Eight Workflows, One Orchestrator

Workflow	From	What it does	Where
Intake agent	Task 4 data preparation	Validates 13 inputs, flags missing optional tests and fills them with the training mode	cdss/schemas.py, features.py
Feature agent	Task 5 feature engineering	Builds the 12 model features in one fixed order from one formula module	cdss/features.py
Risk model	Task 6 machine learning	XGBoost probability of disease (primary verdict)	artifacts/xgb.json
Second-opinion model	Task 7 deep learning	Dense 64-32-16 DNN; flags disagreement with XGBoost	artifacts/dnn.json
Clinical rules agent	Tasks 1-2 knowledge + rules	Rules R1-R7, three-valued: fired / not fired / unknown	cdss/rules.py
Test planner	Task 3 state-space search	A* picks the cheapest tests that make the rules decisive; ML sensitivity ranks them	cdss/advisor.py
Explainer	Task 8 SHAP	Exact per-feature contributions, grouped into body-system domains	cdss/engine.py
Trust agent	Task 8 trust score	Confidence band from validation accuracy plus a 90% conformal set	cdss/engine.py
What-if agent	Task 8 counterfactuals	Model sensitivity over plausible changes; curves for key measurements	cdss/orchestrator.py
Synthesizer	Task 9 integration	Combines everything into the overview, doctor questions and report	cdss/orchestrator.py

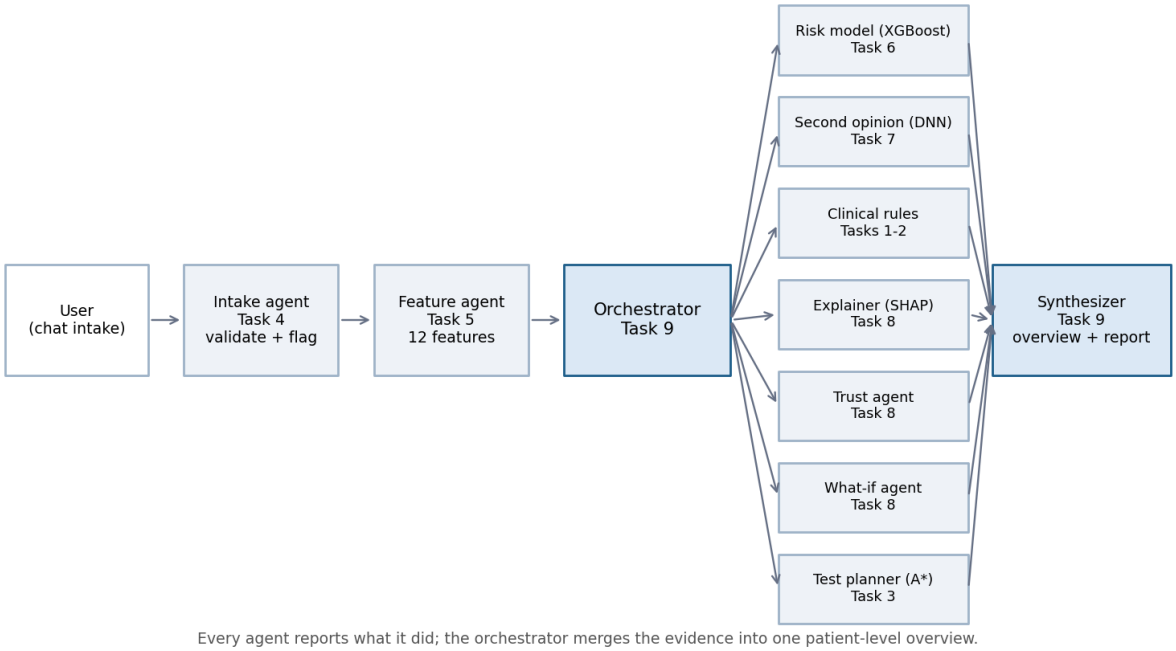


Figure 1. Orchestrated architecture. Each box is an independent workflow that reports what it did.

3. Implementation

3.1 Input validation and missing-data policy

- Pydantic schema with typed ranges (e.g. age 18-100, cholesterol 100-600 mg/dl). Bad input returns field-level errors; it never reaches a model.
- Zero cholesterol or blood pressure is rejected (disguised missing values in the training data).
- Optional tests (ST slope, vessel count, thallium result) may be missing. They are filled with the training mode, flagged, and the confidence band cannot be 'high' when two or more are missing. Rules are never evaluated on guessed values.

3.2 Feature engineering and models

The 12 features follow context.md exactly: cp_asymptomatic, exang, oldpeak_slope_risk, cardiac_risk_score, thalch, cp_atypical angina, oldpeak, chol, hr_reserve, rate_pressure_product, age, chol_age_ratio. Feature formulas live in one function that a parity test compares with the saved feature matrix (max difference under 1e-9), so training and serving cannot drift apart.

Data: 920 raw rows from four hospital sites, 918 after cleaning, prevalence 55.3%. Stratified split 70/15/15 (train 642, validation 138, test 138), seed 42. Imputer, scaler and models are fit on the training split only.

Model	5-fold CV AUC (train)	Test accuracy	Test F1	Test AUC
XGBoost(tuned)	0.876 +/- 0.031	0.819	0.837	0.896
LogisticRegression	0.874 +/- 0.033	0.833	0.852	0.904
SVM	0.859 +/- 0.027	0.819	0.834	0.879
RandomForest	0.853 +/- 0.034	0.841	0.853	0.870
GradientBoosting	0.853 +/- 0.031	0.833	0.854	0.893
XGBoost(context)	0.844 +/- 0.043	0.833	0.852	0.878
DecisionTree	0.811 +/- 0.025	0.819	0.847	0.855

XGBoost was selected by cross-validated AUC on the training split ({"colsample_bytree": 0.7, "eval_metric": "logloss", "learning_rate": 0.03, "max_depth": 2, "min_child_weight": 3, "n_estimators": 200, "random_state": 42, "reg_lambda": 2.0, "subsample": 0.8}). The DNN

(Dense 64-32-16, BatchNorm, Dropout, Adam 1e-3, early stopping) stopped at epoch 84. It is trained in PyTorch and served in plain NumPy from exported weights.

3.3 Explainability, trust and recommendations

- **Explanation:** exact TreeSHAP contributions per patient, summed into four domains (chest symptoms, stress test, cholesterol, overall profile) and shown as plain-language drivers.
- **Trust:** confidence band labelled from validation accuracy per confidence range, a 90% conformal prediction set, and disagreement flags (XGBoost vs DNN, ML vs rules).
- **What-if:** model sensitivity to plausible changes in cholesterol and blood pressure (capped at +/-30%), presented as model behaviour, not treatment advice.
- **Next-test advice:** A* over the missing tests (costs: slope 4, thallium 6, vessel count 8) plus a ranking by whether a result could flip the verdict.
- **Chat:** rule-based, no external AI service. It parses a free-text opening and answers follow-ups only from the computed analysis, so it cannot invent facts.

3.4 User interface

The user view is a guided chat: five short steps, plain-language questions, and a tooltip on every field explaining what it is and why it matters. Chest pain is classified from three yes/no questions rather than by asking for a medical term. The result page leads with a risk pattern, the estimated chance, confidence and information completeness; details sit in four tabs. A separate 'How it works' view serves reviewers.

The flow has two stages. After a red-flag safety screen, stage 1 (home checkup) needs only history, blood pressure, blood tests and a resting ECG and shows a risk tier, never a percentage. If stress-test results exist they can be added for the full multi-workflow analysis (stage 2). With explicit consent the overview can be sent to a demo clinician queue, ranked by urgency. The module is reached from a fictional demo clinic website (folder clinic/): the clinic site is the front door, its button redirects into the app, and the app links back to the clinic and to its booking page. python run_demo.py starts both.

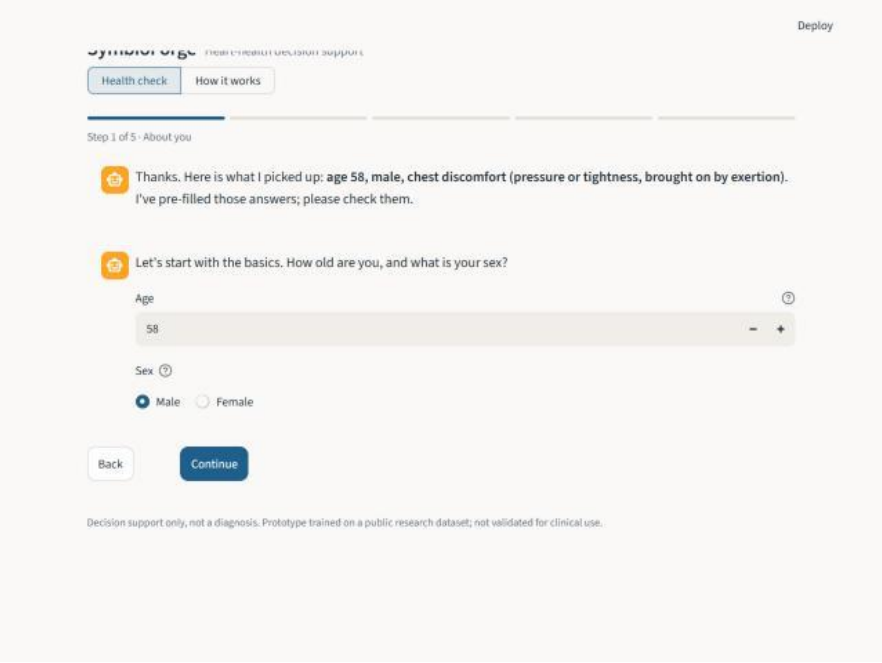


Figure 2. Guided chat intake with free-text pre-fill.

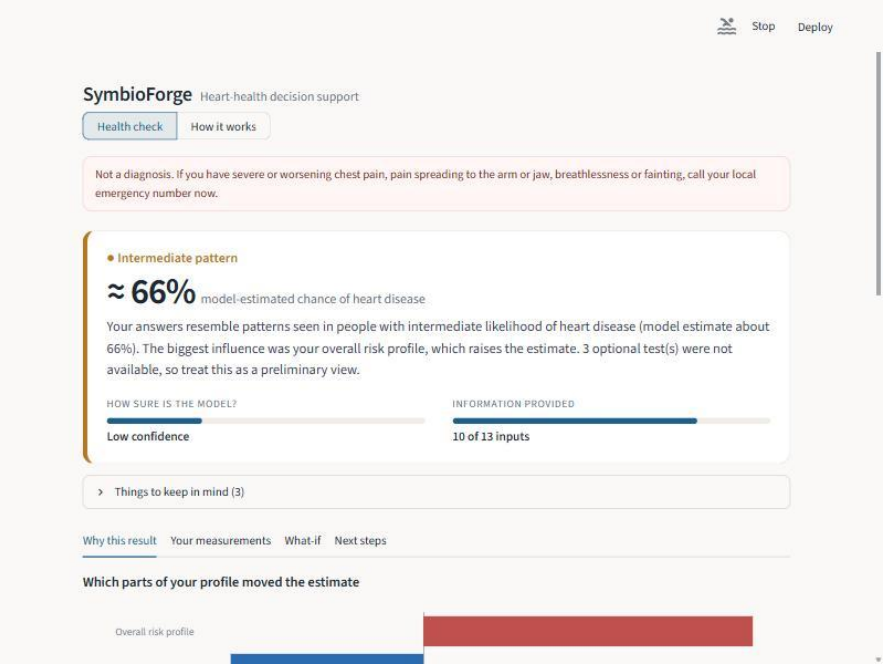


Figure 3. Overview: risk pattern, estimate, confidence and information completeness.

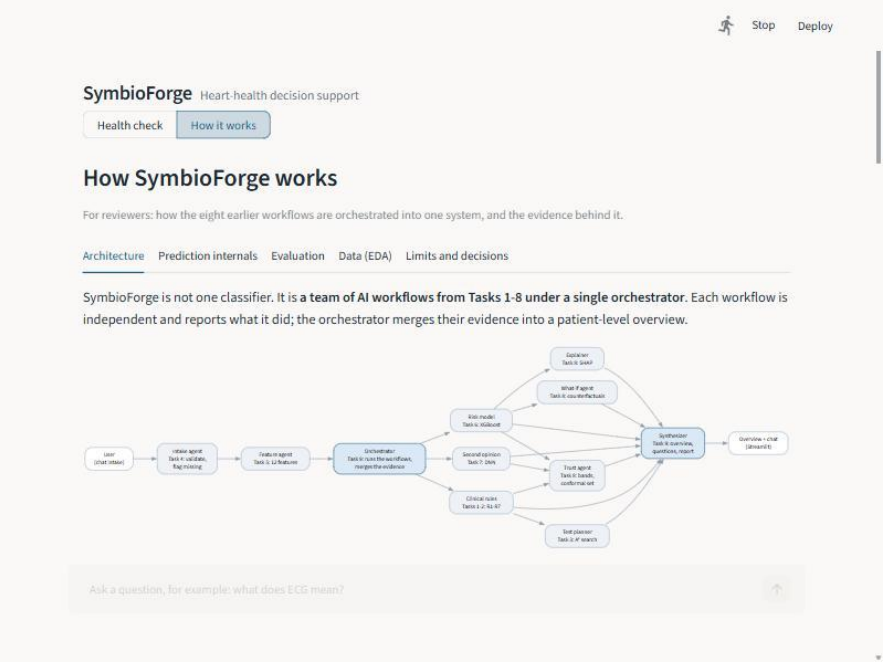


Figure 4. 'How it works' view: orchestration of the workflows for reviewers.

4. Evaluation Report

All figures are on the held-out test split (n = 138), untouched during training, tuning and calibration.

Model	Accuracy	Precision	Recall	ROC-AUC	Brier
xgb	0.819	0.831	0.842	0.896	0.124
dnn	0.819	0.849	0.816	0.897	0.125
mean_ensemble	0.826	0.842	0.842	0.897	0.123

AUC 95% bootstrap interval: XGBoost 0.84-0.95, DNN 0.84-0.95. The intervals overlap heavily, so the models cannot be separated statistically at this sample size. Averaging the two gave no meaningful gain (ensemble AUC 0.897), which is why XGBoost alone gives the verdict and the DNN acts as a second opinion. Always-predict-prevalence Brier baseline: 0.247.

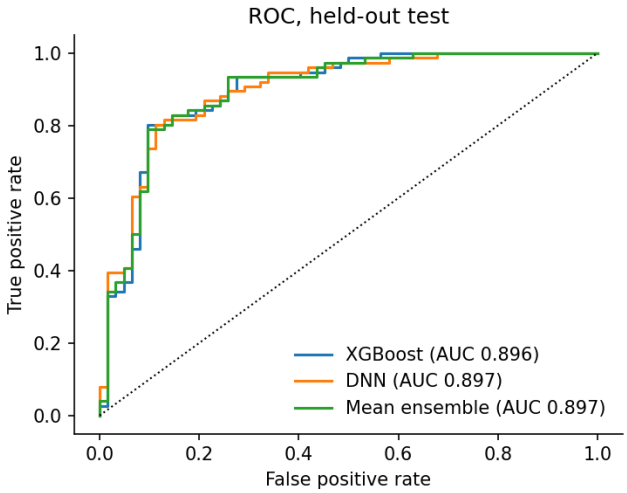


Figure 5. ROC curves, held-out test.

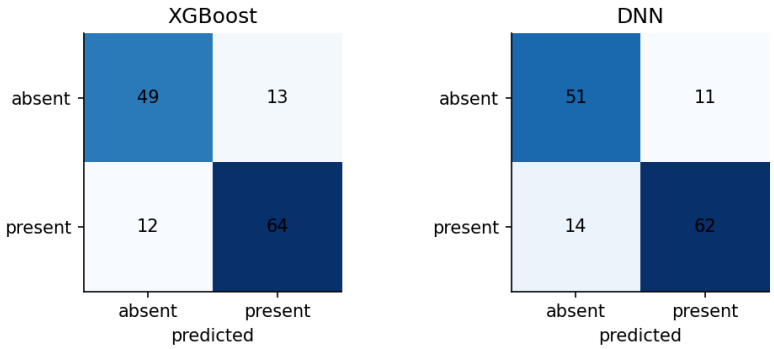


Figure 6. Confusion matrices at the 0.5 threshold.

4.1 Calibration, confidence and conformal prediction

Model confidence range	Test patients	Accuracy
0.50-0.65	23	0.391
0.65-0.75	20	0.950
0.75-0.85	32	0.844
0.85-0.95	50	0.920
0.95-1.00	13	0.923

Accuracy is clearly lowest in the least-confident band (0.391) and between 0.84 and 0.95 in the others. It is not strictly increasing, which is expected with such small bands, but it is why low-confidence results carry a 'defer to clinical judgement' warning. The 90% conformal set covered 92.0% of test patients (target 90%); 79% of patients received a single-label set and 21% the 'uncertain' two-label set. Bands contain only 13-50 patients each, so these figures are approximate.

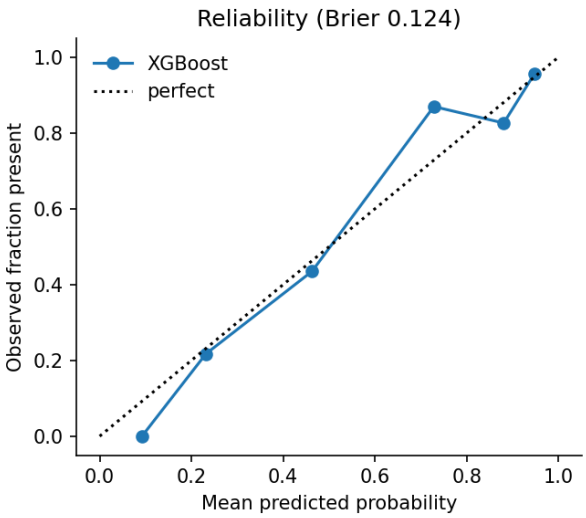


Figure 7. Reliability diagram for XGBoost.

4.2 Cross-checks: rule base and second model

The rule base reaches a decision for 59% of patients (the rest lack a needed test). Where it decides, ML accuracy is 0.921 when rules and ML agree and 0.526 when they disagree (n = 19). XGBoost and the DNN disagree on 6% of patients; XGBoost accuracy is 0.838 when they agree and 0.500 when they do not. Disagreement is therefore a useful warning, and the system lowers confidence when it occurs.

4.3 Effect of missing optional tests (ablation)

Subset	n	Accuracy	AUC
0 imputed	45	0.800	0.895
1 imputed	12	0.917	1.000
2+ imputed	81	0.815	0.884
all optional fields neutralised	138	0.819	0.895

Hiding the three optional tests from every test patient changes AUC only slightly, so the model leans mainly on symptoms and risk factors. The reliability warning is still kept, because the filled-in value can be wrong for an individual. Small subsets, so treat as indicative.

4.4 Explainability validation

Global SHAP ranks: cp_asymptomatic (0.61), cardiac_risk_score (0.51), oldpeak_slope_risk (0.22), oldpeak (0.19), exang (0.19). The dominant role of asymptomatic chest pain agrees with Tasks 1, 2, 5 and 8 (four independent methods). In the training data a 'silent' presentation was more often found to have disease, so the interface warns that no chest pain is not reassuring.

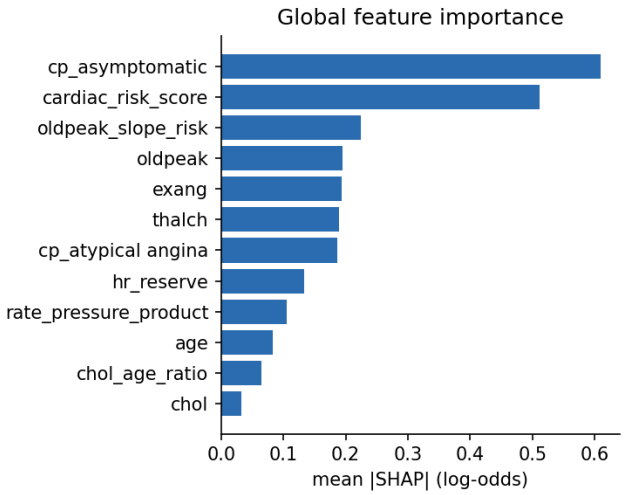


Figure 8. Global feature importance (mean |SHAP|, test set).

4.5 Stage-1 (home checkup) model and the base-rate problem

The full model needs stress-test results that nobody has at home, and it was trained on patients already referred for heart tests (disease rate 55%). A second, 'stage-1' model uses only history, blood pressure, blood tests and a resting ECG. On the same test split it reaches AUC 0.868 (full model 0.896), so little discrimination is lost.

Stage-1 tier (probability thresholds 0.30 / 0.60)	Test patients	Share with disease
lower	34	3%
mid	28	39%
higher	76	84%

Disease prevalence in the people screened	Flags that are right, 'worth a check-up' or above (sens 99%, spec 53%)	Flags that are right, 'see a clinician' (sens 84%, spec 81%)
55%	72%	84%
20%	35%	52%
10%	19%	33%
5%	10%	19%
2%	4%	8%

This is why stage 1 never shows a percentage. The training population has a 55% disease rate; visitors to a clinic website will be far lower, and at 5-10% most flags would be false alarms even though the model discriminates well. Stage 1 is therefore worded as a screening flag that leads to a clinic check, not a finding. The tier groups contain few patients (34 / 28 / 76), so their disease rates are approximate.

4.6 Outlier and edge-case validation

Before claiming anything, behaviour on unusual inputs was tested three ways (script validate_outliers.py).

- **Out-of-fold accuracy on outlier rows:** 64 rows have a value beyond 1.5 x IQR (blood pressure, cholesterol, ST depression). For the full model, AUC is 0.77 on these rows against 0.88 on the rest (accuracy 0.75 vs 0.80); the stage-1 model shows the same pattern (0.73 vs 0.84). The groups are small, but the direction supports not trusting estimates on unusual values.
- **Sweeps across valid ranges:** 32 sweeps of age, blood pressure, cholesterol, peak heart rate and ST depression on four anchor patients. No step jumped by more than 0.17, so the models are smooth. But 5 sweeps moved against the expected clinical direction (blood pressure, cholesterol and, on one anchor, age; swings of only 0.01 to 0.10). In this historical data those two have weak or inverted effects, so the what-if curves now carry a warning and 'actionable' suggestions only ever lower a value.
- **Extreme corners:** all 128 combinations of minimum and maximum valid inputs produced finite outputs; the plausibility guard withheld the estimate for every one of them. On the real dataset it withholds for about 2% of patients.

Mitigation implemented: a plausibility guard (cdss/plausibility.py). A peak heart rate far above the age-expected maximum, very low blood pressure or extreme cholesterol raise a hard flag and cap confidence at 'low'; a value outside the 1st-99th percentile of training data caps confidence at 'moderate'; two or more such values (or any hard flag, or any value beyond the training minimum and maximum) withhold the estimate and show the guideline measurement cards instead. Every probability output (opinion cards, report, chat, what-if curves) is gated on this guard.

5. Testing Report

Area	What is checked	Result
Feature parity	Pipeline output equals the saved feature matrix (max diff < 1e-9); formulas hand-checked	Pass
Model parity	NumPy DNN reproduces the trained DNN's test AUC; XGBoost reproduces its reported AUC (+/- 0.005)	Pass
Rule engine	Three-valued logic, strict/inclusive boundaries (e.g. oldpeak > 2.8, age in [56,58]), R7 default	Pass
Validation	Out-of-range, wrong-type and zero-value inputs return field-level errors	Pass
End-to-end	All four scripted patients (higher risk, lower risk, borderline, rule-vs-ML disagreement) run through the full pipeline	Pass
Determinism	Same input gives identical output, including the counterfactual	Pass
A* planner	Returns the cheapest decisive test set (slope + vessel count, cost 12) for the borderline case	Pass
Cleaning	Zero-to-missing, duplicate removal, target binarisation, no constant filler for imputed cholesterol	Pass
Chat and glossary	Intro parsing, status cut-offs, every suggested question answered, emergency wording	Pass
Safety and guard	Red-flag symptoms end the flow before any model runs; implausible or out-of-range values withhold the estimate; stage 1 never shows a percentage	Pass
Clinic queue	Nothing stored without consent; urgent cases sorted first; disagreement bumps a case up; counterfactuals only lower modifiable factors	Pass
Dashboard	Streamlit test runs the guided flow and every example through to the result page, and the How-it-works view	Pass

31 automated tests pass. Robustness: invalid inputs rejected 8/8; 300 random valid patients processed without error (300/300). Latency of the full pipeline including SHAP, counterfactual and planner: median 115 ms, 95th percentile 191 ms (target under 500 ms). Two independent code reviews found 15 defects. The first found six (a constant imputation value that acted as a hidden disease marker, miscalibrated confidence labels, an ambiguous headline probability, a spec mismatch, misleading 'actionable' counterfactuals and rule statistics that used imputed values); five were fixed and one was recorded as a deliberate design change. The second found nine more in the newest code (bypasses of the guard and of the red-flag screen, probabilities leaking when an estimate was withheld, queue ordering and migration, an open clinician view, a guard that failed open, and stage-1 behaviour driven by referral bias); all nine were fixed.

Domain and user validation: scripted cases and automated checks only. No clinician or end-user study has been carried out; this is listed as future work.

6. Responsible AI Assessment

Area	Finding and mitigation
Intended use	Educational decision support for one patient at a time. Emergency and not-a-diagnosis notices appear on every screen and in the report.
Data	Public research data from four hospitals, collected decades ago. Not representative of any current population.
Safety screen	Red-flag symptoms (severe pain now, pain spreading, breathlessness at rest, fainting, cold sweat) stop the flow with an emergency message before any model runs. Examples and 'Edit my answers' cannot skip it.
Base rate	Training data are referred patients (55% disease). Stage 1 therefore shows a tier only and says it is a screening flag; Section 4.5 quantifies the false-alarm rate in lower-prevalence populations.
Outliers	Estimates are withheld for implausible values, values beyond the training minimum or maximum, or two edge values (about 2% of real patients); outlier rows were less accurate out-of-fold (Section 4.6).
Consent and data	Nothing is stored unless the user ticks consent and sends the overview; the demo queue uses fictional data and a shared demo passcode instead of real login. A real deployment would need authentication, encryption, audit logs and compliance (regulated medical-device software; health-data law such as India's DPDP Act).

Missing data	Flagged, never silent. Two or more filled fields block a 'high' confidence band. Rules never run on guessed values.
Out-of-distribution	Age outside the 28-77 training range triggers a warning.
Transparency	SHAP per patient, plain-language reasons, rule trace, agreement chips, a visible trace of every workflow. The DNN has no per-patient explanation, and the UI says so.
Uncertainty	Validation-based confidence bands, conformal set, and disagreement flags. 'Uncertain' is an allowed output.
Counterfactuals	Labelled model sensitivity, restricted to cholesterol and blood pressure, capped at +/-30%; never presented as treatment effect.
Chat	Rule-based and offline; answers are assembled from computed evidence and cannot hallucinate. Emergency wording is always available.
Privacy	Stateless: no patient data stored or logged; no data leaves the machine; no external AI service.
Reproducibility	Fixed seeds, pinned split, saved artifacts, determinism test, one-command rebuild.

Fairness check (test set). Groups are small, so differences are weak evidence.

Group	n	AUC	Recall	Precision	Prevalence
sex: male	113	0.881	0.843	0.843	0.62
sex: female	25	0.895	0.833	0.714	0.24
age_band: >65	13	0.955	0.909	0.909	0.85
age_band: 56-65	47	0.802	0.871	0.794	0.66
age_band: 46-55	51	0.911	0.800	0.833	0.49
age_band: <=45	27	0.981	0.778	0.875	0.33
site: Cleveland	46	0.898	0.783	0.818	0.50
site: Hungary	53	0.897	0.800	0.762	0.38
site: Switzerland	18	0.941	0.882	1.000	0.94
site: VA Long Beach	21	0.825	0.938	0.789	0.76

Limitations: women are a small share of the data; disease prevalence varies strongly by hospital (for example 94% in one site against 36% in another), and hospital also drives which values are missing, so it acts as a hidden confounder; the test set is small; there is no external or temporal validation.

7. Requirements and Design Decisions

ID	Requirement
FR1-FR2	Validate raw inputs; handle optional/missing inputs explicitly
FR3-FR5	Build the 12-feature vector exactly; run XGBoost and DNN; verdict from XGBoost with disagreement flag
FR6	Evaluate rule base R1-R7, report fired/unknown rules and ML agreement
FR7-FR9	SHAP explanation; confidence band with reliability warnings; counterfactual sensitivity
FR10	Next-test recommendation by A* when key tests are missing
FR11-FR12	Readable summary and downloadable report; guided UI from inputs to overview
NFR	Latency under 500 ms (met), deterministic, stateless, one-command run, zero cost, every feature traceable to its formula

Key design decisions (full log in ARCHITECTURE.md):

Decision	Why
XGBoost gives the verdict; the DNN is a second opinion	The mean ensemble was no better, and SHAP explains XGBoost, keeping verdict and explanation consistent
Three-valued rules	A rule must not fire on a guessed value; the unknowns drive the test planner
Random draw instead of a constant for unobservable cholesterol	A constant filler became a hidden site marker the model learned (found in review)
DNN trained in PyTorch, served in NumPy	TensorFlow would not load on the development machine; no heavy runtime dependency
Confidence labels from validation accuracy	Fixed probability cut-offs ignored calibration
Rule-based chat, no external AI service	Zero cost, offline, cannot invent facts; trade-off is limited open-ended understanding

8. User Manual and Deployment Guide

8.1 Using the system

- Open the app. 'Health check' is for users; 'How it works' is for reviewers.
- Describe symptoms in your own words (optional) or press Start. Answer the five short steps; hover the '?' next to any field for a plain explanation.
- Read the overview: risk pattern, estimated chance, confidence and how much information was provided.
- Use the tabs: Why this result, Your measurements, What-if, Next steps (including questions for a doctor and a downloadable report).
- Ask follow-up questions in the chat, for example 'What does thallium mean?'. For a quick demo use the four ready-made examples on the first screen.

8.2 Running and deploying

Local run (PowerShell):

```
cd C:\Users\aarya\OneDrive\Desktop\CODING\SymbioForge
pip install -r requirements.txt
streamlit run app.py
```

Rebuild everything: pip install -r requirements-train.txt, then python train.py, python eda.py, python evaluate.py, python build_docs.py, python -m pytest. Cloud deployment: push the repository to GitHub and create a Streamlit Community Cloud app with main file app.py (Python 3.11+, requirements.txt only; no secrets and no database). Artifacts are under 300 KB and committed, and XGBoost is stored as portable JSON so it does not depend on the training machine.

Deployment readiness: all dependencies are free and open source; the app starts cold in about 15 seconds on a slow disk; no external services are required.

9. Deliverables Checklist

Task	Deliverable	Where
9	Application (running instance)	streamlit run app.py
9	Integrated source code	cdss/, ui/, app.py, train.py, evaluate.py, eda.py
9	Dashboard	Health check view (two stages), How it works view, clinician queue (demo)
9	Clinic integration demo	clinic/ (static demo clinic site that redirects into the app); run_demo.py
9	Report generator	cdss/report.py; download button; reports/sample_report.html
9	Deployment documentation	README.md and Section 8 of this report
10	Evaluation report	Section 4; reports/evaluation.json
10	Responsible AI report	Section 6
10	User manual	Section 8.1
10	SRS / SDD	Section 7, SPEC.md, ARCHITECTURE.md
10	Testing report	Section 5; tests/test_core.py
10	Final project report	This document
10	Deployment guide	Section 8.2

10. Conclusion and Future Enhancements

The experiments are integrated into one system whose value lies in the combination: eight workflows produce independent evidence, an orchestrator reconciles them, and the patient receives an explained, uncertainty-aware overview rather than a bare label. On the held-out test split XGBoost reaches 81.9% accuracy and 0.896 AUC, the system answers in about 115 ms,

and all 31 automated tests pass. The prototype is honest about its limits: small test set, historical single-source data, no clinical validation.

Future work: repeated cross-validation and site-held-out validation; calibrated probabilities (isotonic); a clinician and user study; an optional language-model layer for the chat, restricted to the computed analysis; and multilingual support.